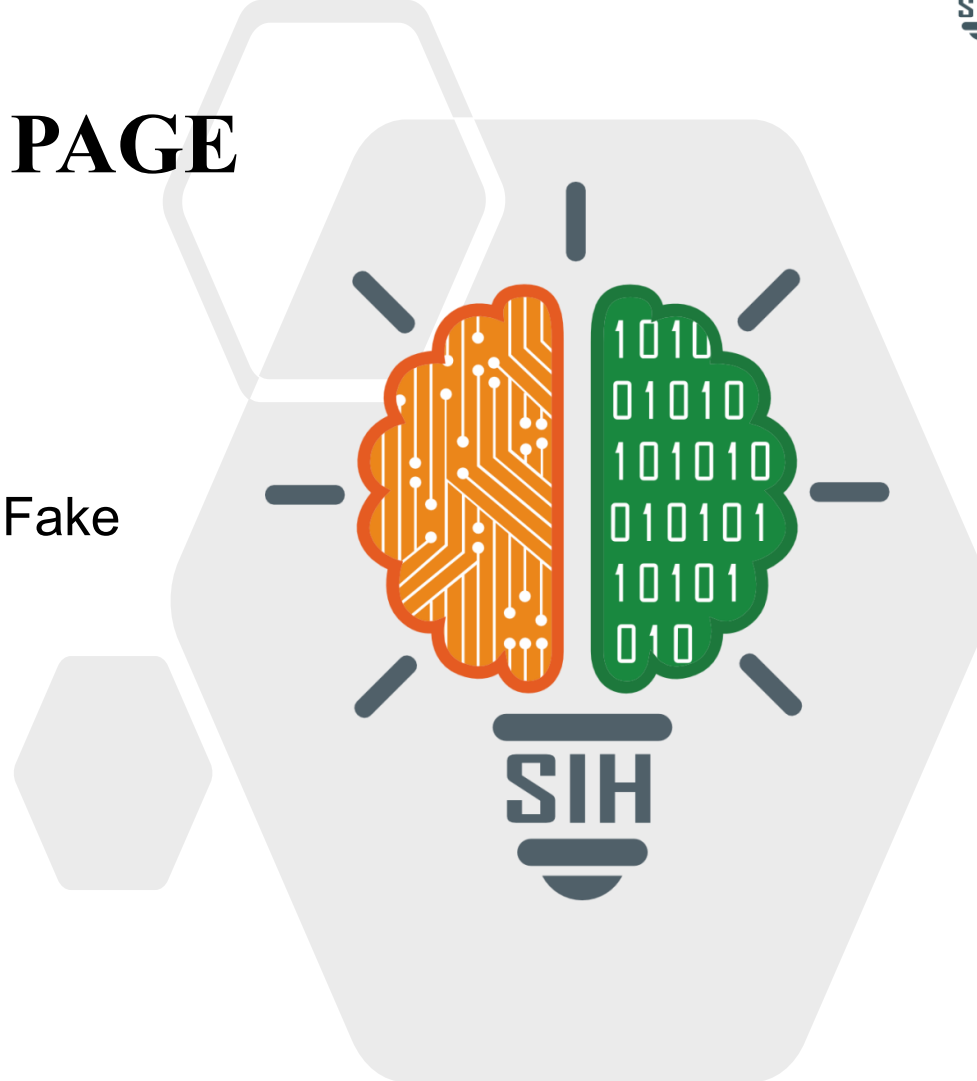


# SMART INDIA HACKATHON 2026



## TITLE PAGE

- **Problem Statement ID** – SIH26188
- **Problem Statement Title** - AI-Based Fake Identity & Document Screening System
- **Theme** – Blockchain & Cybersecurity
- **PS Category** - Software
- **Team ID** -
- **Team Name** - ZYNTRIX



# IDEA TITLE : SENTINELID

## AI-Based Fake Identity & Document Screening System

### ❑ Problem:

- Sophisticated document forgery poses critical security risks at land-border checkpoints that rely on manual verification.

### ❑ Our Idea:

- **SentinelID** is an **AI-powered border document screening platform** that validates documents instantly against official databases.
- It replaces manual guesswork **with multi-layered forensic analysis**, identifying physical and digital tampering invisibly

➤ **DEMO: [LINK](#)**



### ❑ Solution:

- Our automated system streamlines identity verification by performing advanced sequence checks on every submitted document.
- The platform scans the ID **using high-accuracy OCR** for database validation, followed by strict forensic tampering detection and AI-driven face verification against a live selfie.
- Ultimately, all findings are consolidated to **output a real-time Risk Score (0-100) complete with explainable evidence highlights** that clearly flag any anomalies for manual reviewers.

### ❑ Innovations/Uniqueness:

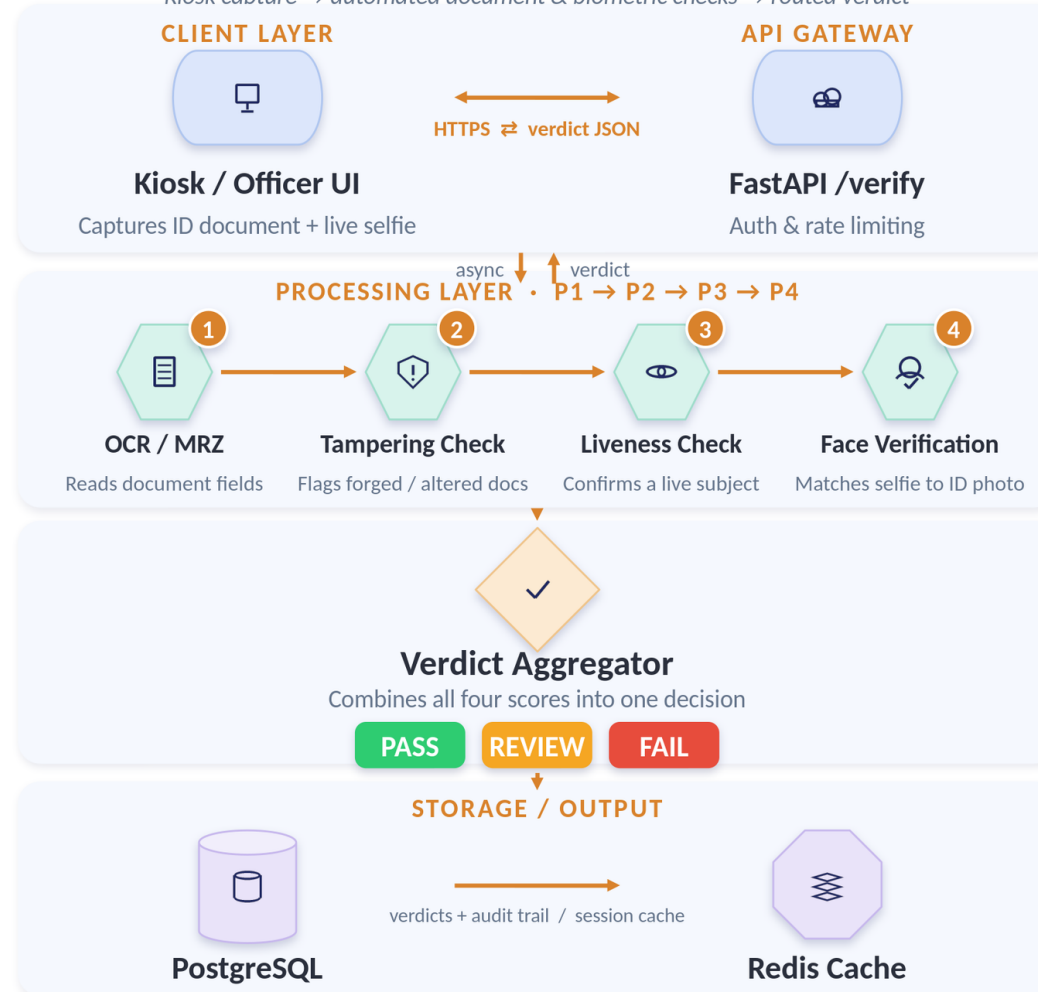
- **Explainable AI overlays:** Grad-CAM / Error Level Analysis heatmaps.
- **Two-factor forensic proof:** visual tampering detection layered with EXIF/metadata and noise analysis.
- **Offline-capable edge inference** for checkpoints with poor connectivity.
- **Fuzzy watchlist matching:** handles transliteration variants of names.
- **Hash-chained audit trail** for investigations, tying into the Blockchain & Cybersecurity theme.

## WORKFLOW PIPELINE



## IDENTITY VERIFICATION PIPELINE

Kiosk capture → automated document & biometric checks → routed verdict



## Technologies Used

- **React** – Used for the **production-level kiosk interface**.
- **FastAPI** – Backend framework for **handling API requests** and coordinating verification processes.
- **PassportEye** – Extracts and parses **passport MRZ information**.
- **PyTesseract** – Performs **OCR** to extract text from documents.
- **MobileNetV2** – **Lightweight CNN** used for image classification and tamper analysis.
- **Anti-Spoofing CNN** – Performs **face liveness detection** to identify real users vs. presentation attacks.
- **DeepFace** – Framework used for **face analysis and verification**.
- **ArcFace** – Generates **face embeddings** and performs face matching.
- **PostgreSQL** – Stores **verification results and audit logs**.
- **Redis** – Provides fast **session and result caching**.
- **Document Scanner/Webcam** – Captures **identity documents and facial images**.
- **NUC-class Edge Box** – **Runs OCR, tamper detection, and liveness processing locally when connectivity is limited**.

## FEASIBILITY ANALYSIS

## TECHNICAL

- Uses open-source, CPU-friendly models with functional Colab prototypes (P1, P2, P4) ready; technical effort focuses only on integration and liveness selection.

## FINANCIAL

- Open-source stack eliminates recurring inference fees, restricting costs to pre-budgeted, one-time edge hardware purchases (CapEx).

## MARKET

- Tailor-made for border checkpoints to directly address SSB's operational problem statement, eliminating market demand uncertainty.

## OPERATIONAL

- Runs offline at edge-first checkpoints to prevent downtime, needing only periodic model threshold updates using deployment data..

## THE CHALLENGES

## LIGHTING &amp; LOW-RES SCANS



Glare, camera blur, or low lighting at checkpoints degrading OCR extraction accuracy.

## NATURAL DOCUMENT WEAR



Normal creasing, folding, or fading incorrectly categorized as deliberate digital tampering.

## REMOTE BORDER CONNECTIVITY



Unstable or non-existent internet access at remote SSB border outposts.

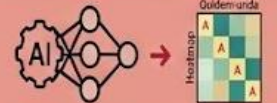
## MITIGATION STRATEGIES

## AUTOMATED PREPROCESSING PIPELINE



Integrated adaptive thresholding, bilateral filtering, and deskewing to restore low-quality captures before model inference.

## CALIBRATED CONFIDENCE THRESHOLDS &amp; HUMAN-IN-THE-LOOP



Calibrated model training using datasets (MIDV-500, Kaggle forged document sets) separating digital manipulation artifacts from physical wear-and-tear.

## EDGE-NATIVE ARCHITECTURE



Fully containerized Docker workflow enabling full offline operation with local hash-chain storage that syncs automatically when network availability resumes.

## Direct Impact on Target Users

- **Instant Decision Support:** Collapses document verification time from minutes down to **double-digit seconds**, drastically cutting **checkpoint processing delay**.
- **Fatigue & Error Reduction:** Replaces stressful manual text-reading and photo-eyeballing with concurrent AI processing, **mitigating fatigue-induced mistakes**.
- **Standardized Inspection:** Delivers a structured **PASS / REVIEW / FAIL** risk score, eliminating subjective guesswork and variance across officers or shifts.
- **Zero-Spoof Protection:** Enforces a mandatory "**Liveness Gate**" that halts verification if a live person is not present, protecting officers from **printed-photo or screen-replay tricks**.

## Strategic Impact

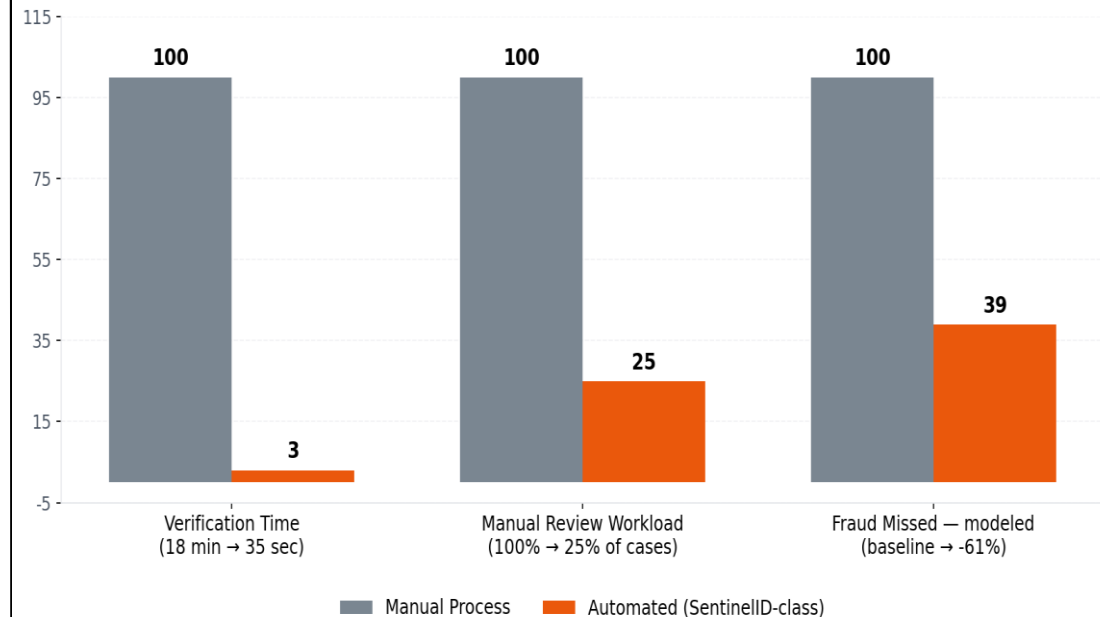
- **Invisible Threat Detection:** Uncovers subtle digital tampering—such as **swapped photos, modified birth dates, or forged stamps**—invisible to the naked eye.
- **Resilient Edge Deployment:** Operates continuously at **low-connectivity border posts** without central network dependence, syncing audit logs automatically when connection restores.
- **Actionable Intelligence Pipeline:** Converts routine, temporary border decisions into **structured digital audit trails** with sub-scores for downstream criminal investigations.

## Economic & Operational Benefits

- **Cost-Effective Scalability:** Built on open-source AI frameworks to **run locally on low-cost edge hardware**, eliminating recurring per-scan API costs.
- **Throughput Optimization:** Maximize checkpoint capacity during **peak traffic hours** without requiring extra personnel or physical infrastructure.
- **Cross-Domain Interoperability:** Provides a flexible verification core easily expandable to broader national uses, including **telecom onboarding and banking KYC checks**.

### Manual vs. Automated Identity Verification — Industry-Benchmarked

Indexed to manual baseline = 100 so three different-unit metrics (minutes, workload %, a modeled index) can share one axis — real units shown in each label below





# RESEARCH AND REFERENCES



## ICAO Standards (MRZ & Identity Architecture):

Reference: International Civil Aviation Organization. (2021). Doc 9303: Machine Readable Travel Documents (8th ed.). ICAO, Montreal, Canada.

[LINK](#)

## Document Image Forensics & ELA:

Reference: Krawetz, N. (2007). A Picture's Worth. Digital Image Analysis and Forensics. Black Hat Briefings.

[LINK](#)

## ArcFace (Biometric Embeddings):

Reference: Deng, J., Guo, J., Xue, N., & Zafeiriou, S. (2019). ArcFace: Additive Angular Margin Loss for DeepFace Recognition. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 4690–4699

[LINK](#)

## DeepFace Framework Implementation:

Reference: Serengil, S. I., & Ozpinar, A. (2021). HyperExtended LightFace: A Facial Attribute Analysis Framework. International Conference on Engineering and Telecommunication (EnT), 1–4.

[LINK](#)

## RetinaFace (Robust Face Detection):

Reference: Deng, J., Guo, J., Ververas, E., Kotsia, I., & Zafeiriou, S. (2020). RetinaFace: Single-Shot Multi-Level Face Localisation in the Wild. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 5203–5212. [LINK](#)

## Smart Engines MIDV-500 Dataset:

Reference: Arlazarov, V. V., Bulatov, K., Chernov, T., & Arlazarov, V. L. (2019). MIDV-500: A Dataset for Identity Documents Analysis and Recognition on Mobile Devices in Video Stream. Computer Optics, 43(5), 818–824.

[LINK](#)